

JSON Exam: Sports Data Management System

Total Points: 20 Duration: 90 minutes

Question 1: Transform XML Sports Data to HTML Using XSLT (10 Points)

Objective:

Write an XSLT stylesheet to transform XML data representing sports events into an HTML table.

XML Data Example:

```
<sports>
  <title>Upcoming Sports Events</title>
  <caption>Latest Scheduled Matches</caption>
  <description>Find information about the latest sports events,
    including teams, venues, and schedules.</description>
  <events>
    <event>
      <id>101</id>
      <name>Champions League Final</name>
      <sport>Soccer</sport>
      <date>2024-06-01</date>
      <venue>Wembley Stadium</venue>
      <status>Scheduled</status>
    </event>
    <event>
      <id>102</id>
      <name>NBA Finals Game 5</name>
      <sport>Basketball</sport>
      <date>2024-06-15</date>
      <venue>Staples Center</venue>
      <status>Completed</status>
    </event>
    <event><id>103</id>
      <name>US Open Tennis Final</name>
```

```
<sport>Tennis</sport>
<date>2024-09-10</date>
<venue>Arthur Ashe Stadium</venue>
<status>Scheduled</status>
</event>
</events>
</sports>
```

Tasks and Scoring:

- **Create the XML Data File:** Ensure the XML file is well-structured and properly formatted. **[1 Point]**
- **Write XSLT to Generate HTML:** Create an XSLT file to transform the XML data into a styled HTML table. **[2 Points]**
- **Display Metadata:** Use the XSLT file to display the title, caption, and description above the event table. **[2 Points]**
- **Render Event Table:** Generate a table from the `<events>` element with columns:
 - Event ID
 - Name
 - Sport
 - Date
 - Venue
 - Status**[3 Points]**
- **Add Conditional Styling:**
 - Highlight rows based on event status:
 - **Green background** for "Scheduled".
 - **Gray background** for "Completed".**[2 Points]**

Question 2: JSON Sports Data Rendering with Interactivity (10 Points)

Objective:

Use JSON data to dynamically render sports events and add interactivity using HTML, CSS, and JavaScript.

JSON Data Example:

```
{
  "events": [
    {
      "id": 201,
      "name": "Super Bowl LVIII",
      "sport": "American Football",
      "date": "2024-02-11",
      "venue": "Allegiant Stadium",
      "status": "Scheduled"
    },
    {
      "id": 202,
      "name": "Wimbledon Men's Final",
      "sport": "Tennis",
      "date": "2024-07-14",
      "venue": "Centre Court",
      "status": "Completed"
    },
    {
      "id": 203,
      "name": "F1 Grand Prix - Monaco",
      "sport": "Formula 1",
      "date": "2024-05-26",
      "venue": "Monaco Circuit",
      "status": "Scheduled"
    }
  ]
}
```

Tasks and Scoring:

- **Create the JSON File:** Ensure the JSON file is valid and formatted correctly. **[1 Point]**
- **Render Event Table:** Use JavaScript to dynamically display the events in a table format with columns:
 - Event ID
 - Name
 - Sport
 - Date
 - Venue
 - Status [3 Points]

- **Implement Sorting:** Add a "**Sort by Date**" button that toggles between ascending and descending order. **[3 Points]**
- **Filter Events by Status:** Add a dropdown to filter events based on status ("All", "Scheduled", "Completed"). **[3 Points]**

Additional Requirements:

- **Responsive Styling:** Use **CSS Grid or Flexbox** to ensure the table is responsive across various screen sizes. **[2 Points]**
- **Highlight Past Events:** If an event's date is **before today's date**, add a **badge** next to the event labeled "**Past Event**". **[3 Points]**

Submission Instructions:

Submit the following files in a **zipped folder** named:

WebExam_Sports_<YourName>.zip

index.html style.css script.js events.json sports_data.xml sports_data.xslt

Ensure your project is tested and functional in a local web environment (e.g., **Live Server**, **XAMPP**, **WAMP**).

This **Sports Management System** covers **XML & XSLT transformation, JSON data handling, dynamic table generation, sorting, filtering, and responsive UI design** using **HTML, CSS, and JavaScript**.